

THE MALAWI NATIONAL EXAMINATIONS BOARD

2002 MALAWI SCHOOL CERTIFICATE OF EDUCATION EXAMINATION

MATHEMATICS

Subject Number: M131/II

Wednesday, 16 October

Time allowed: 2 h 30 mins
1:30 - 4:00 pm

PAPER II (100 marks)

Instructions:

1. This paper contains 6 pages. Please check.
2. Answer **all** the **six** questions in Section **A** and any **three** questions from Section **B**.
3. The maximum number of marks for each answer is indicated against each question.
4. Mathematical tables, graph paper and answer books are provided.
5. Used graph paper and/or supplementary sheets must be tied together inside the answer book with a string.
6. All working must be clearly shown: it should be done on the same sheet as the rest of the answers.
7. Use of electronic calculators is **not** allowed in this paper.
8. Write your **Examination Number** on top of each page of your Answer Book.

Section A (55 marks)

Answer all the six questions in this section.

1. a. Simplify $\frac{3\frac{1}{7} \left[\left(2\frac{1}{4}\right)^2 - 3\frac{3}{4} \right]}{3\frac{7}{16}} = \frac{\frac{22}{7} \left\{ \left(\frac{9}{2}\right)^2 - \frac{15}{1} \right\}}{\frac{25}{16}} = \frac{\frac{22}{7} \cdot \frac{81-60}{4}}{\frac{25}{16}} = \frac{\frac{22}{7} \cdot \frac{21}{4}}{\frac{25}{16}} = \frac{6}{5}$ (4 marks)

b. Solve for x , $\log_3 x - 2 \log_3 x = 2$. $-\log_3 x = 2 \Rightarrow x = \frac{1}{9}$ (4 marks)

2. a. Simplify $\frac{(\sqrt{24} - \sqrt{6})^2}{\sqrt{6}}$ leaving your answer with a rational denominator.

$$\frac{(2\sqrt{6} - \sqrt{6})^2}{\sqrt{6}} = \frac{(\sqrt{6})^2}{\sqrt{6}} = \sqrt{6}$$

b. Make d the subject of the formula. $s = \frac{n}{2} [2a + (n-1)d]$ (3 marks)

3. a. Express as a single fraction

$$\frac{4}{x-5} - \frac{5}{x(x-5)} - \frac{3}{x}$$

$$\frac{2x}{n} = 2a + (n-1)d$$

$$\frac{2x}{n} - 2a = (n-1)d$$

$$d = \frac{1}{n-1} \left(\frac{2x}{n} - 2a \right)$$

$$\frac{4(-5-3(-5))}{x(x-5)} = \frac{4(-5-3(-5))}{x(x-5)} = \frac{4(-5+15)}{x(x-5)} = \frac{4(10)}{x(x-5)} = \frac{40}{x(x-5)}$$

b. Evaluate $\sqrt[3]{\frac{\tan 38^\circ 34'}{35.71}}$ using logarithm tables, leaving your answer to 3 decimal places. (3 marks)

4. a. Using a ruler and a pair of compasses only, construct in the same diagram:

(i) a triangle ABC with base $BC = 12$ cm, $AB = 10$ cm and angle $ABC = 45^\circ$;

(ii) a perpendicular from A to BC , meeting BC at N ;

(iii) a circle which touches BC at N and also touches AC . Measure the radius of this circle. (7 marks)

b. In Figure 1, XYZ is a tangent at Y on a circle $YSTR$.

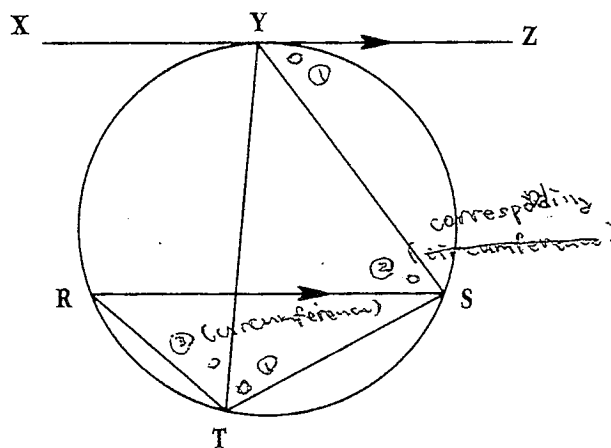


Figure 1

If XZ is parallel to RS , prove that YT bisects angle RTS .

(4 marks)

Continued →

- (5 marks)

- $$23 \left(\times \frac{56}{2} \right)^2 \cdot \cancel{\pi} \neq$$
-
- $$= \cancel{\pi} \left(\times 2 \times \right) \left\{ \left(\frac{28}{2} \right)^2 \cancel{\pi} - \left(\frac{14}{2} \right)^2 \cancel{\pi} \right\}$$

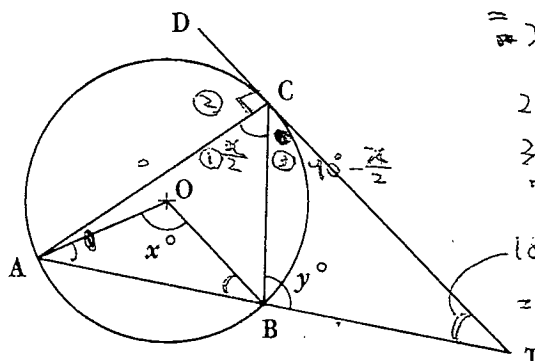


Figure 2

(4 marks)

- (3 marks)

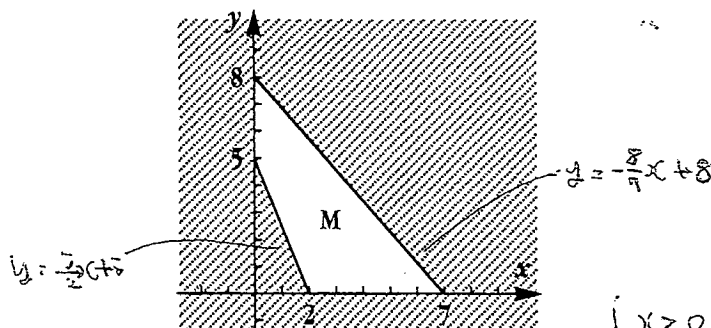


Figure 3

Write down the four inequalities describing the region M .

- $$\begin{cases} x > 0 \\ y > 0 \\ y < -\frac{8}{7}x + 8 \\ y > -\frac{5}{2}x + 5 \end{cases} \quad (5 \text{ marks})$$

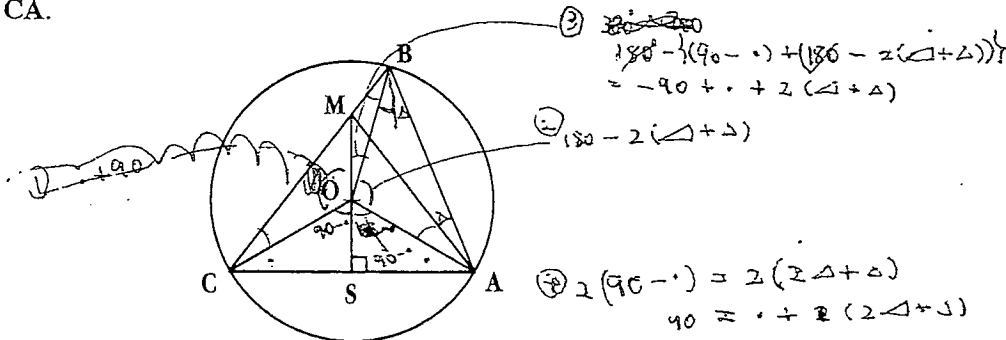


Figure 4

Prove that:

- (i) the triangles MCS and MAS are congruent;
(ii) angle MCO = angle MAO;
(iii) MBOA is a cyclic quadrilateral.

$$\{180 - 2(\Delta + \Delta)\} + \{-10 + 4(2\Delta - \Delta)\} + 2\Delta + 0$$

(6 marks)

Continued

Section B (45 marks)

Answer any **three** questions from this section.

7. The worker finds that the earnings per week depend on the time spent in the shop and in the office. If n hours per day are spent in the shop, the weekly earning p kwacha, are given by the relation, $p = 11 + 24n - 3n^2$.

$$\begin{array}{r} 131 \\ 75 \\ \hline 106 \end{array}$$

- a. (i) Copy and complete the following table of values for $p = 11 + 24n - 3n^2$.

n	0	1	2	3	4	5	6	7	8
p	11	32	47	56			47	32	11

$$P(4) = 11 + 96 - 48 = 59$$

$$P(5) = 11 + 120 - 75 = 106$$

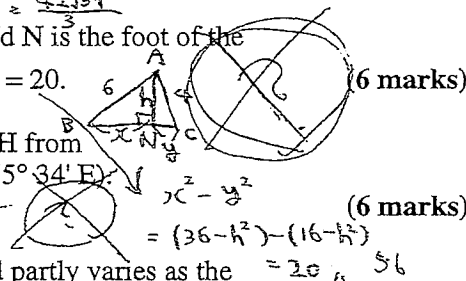
- (ii) Using a scale of 2 cm to represent 10 units on the vertical axis and 2 cm to represent 1 unit on the horizontal axis, draw the graph of $p = 11 + 24n - 3n^2$.

- (iii) Use your graph to find the possible times that the worker may stay in the shop to earn K40.00.

$$40 = 11 + 24n - 3n^2 \quad n = \frac{12 \pm \sqrt{144 - 87}}{3} = \frac{12 \pm \sqrt{57}}{3} \quad (9 \text{ marks})$$

$$3n^2 - 24n + 29 = 0$$

- b. In the acute-angled triangle ABC, AB = 6 cm, AC = 4 cm and N is the foot of the perpendicular from A to the side BC. Show that $BN^2 - NC^2 = 20$.



8. a. Town X is 10 km due north of town Y. The bearing of ship H from town X is $145^\circ 34'$ (S $34^\circ 26'$ E) and town Y is $055^\circ 34'$ (N $55^\circ 34'$ E). How far is the ship from Y?

- b. The time (T) it takes to enter a stadium is partly constant and partly varies as the number of people (N) entering the stadium. If there are 4 people it takes 12 seconds to enter the stadium and if there are 5 people it takes 14 seconds. How long will it take, if there are 28 people entering the stadium.

$$T = aN + b$$

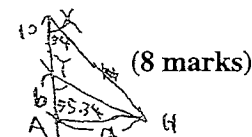
$$\begin{aligned} 12 &= 4a + b \\ 14 &= 5a + b \end{aligned} \quad \begin{aligned} a &= 2 \\ b &= 4 \end{aligned} \quad (9 \text{ marks})$$

9. a. Using a ruler and a pair of compasses only, construct in the same diagram:

- (i) a triangle LMN in which angle MLN = 60° , LM = 7.5 cm and LN = 5.0 cm;

- (ii) the point R on MN such that MR : RN = 2 : 1;

- (iii) Measure and state the length of LR.



- b. Figure 6 shows the graph of $y = (x+2)(x-1)(x-3)$.

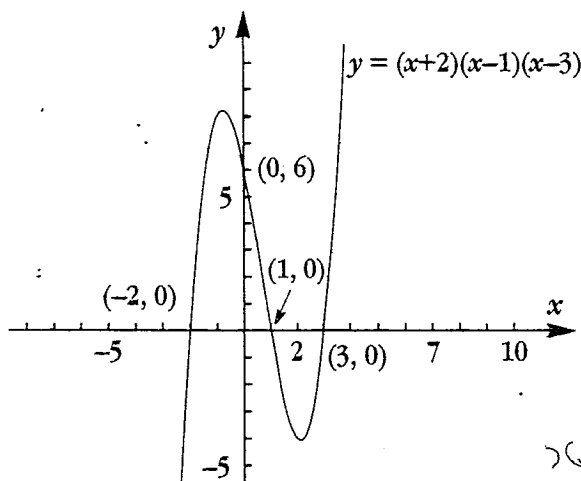


Figure 6

Use the graph to find the solutions to the following equation $x^3 - 2x^2 - 6x + 6 = 0$.

(7 marks)

Continued →

10. a. The sum of the first three positive numbers which are in a GP is 52. The square of the second number is equal to four times the third number. Find the second term of the progression. (10 marks)
- b. In Figure 7, $PR = PS = 13$ cm, $QR = QS = 10$ cm and $RS = 8$ cm.

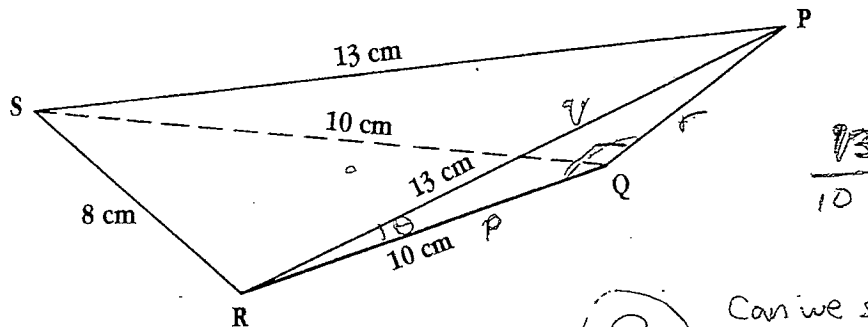


Figure 7

Calculate the angle between the face PRS and base RQS.

(5 marks)

11. a. An employee works as a mechanic and a minibus driver at a company. The terms of employment are listed below.

- to work for a maximum of 40 hours per week.
- to spend at least 16 hours per week mending cars and at least 5 hours per week driving a minibus.
- to spend at least twice as much time mending cars and driving a minibus.

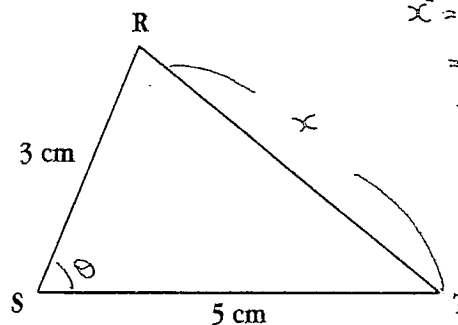
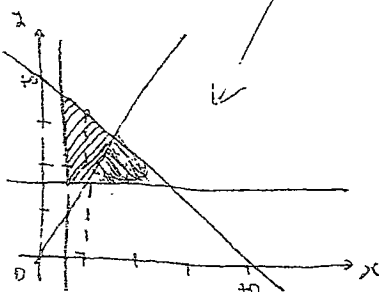
- (i) Express the above conditions as inequalities, using y to represent the number of hours spent mending cars and x to represent number of hours spent driving minibus.
- (ii) Using a scale of 2 cm to represent 5 hours on y -axis and 2 cm to represent 5 hours on the x -axis draw graphs of the inequalities and shade the unwanted region.
- (iii) If the employee spends 10 hours on driving, use the graph to find the maximum number of hours that can be spent on mending cars.

$$\begin{cases} x + y \leq 40 \\ y \geq 16 \\ x \geq 5 \\ y \geq 2x \end{cases}$$

$$y = -10 + 40 = 30 //$$

(10 marks)

- b. In Figure 8, triangle RST is such that $RS = 3$ cm, $ST = 5$ cm and Cosine of angle RST = 0.600.



$$\begin{aligned} x^2 &= 3^2 + 5^2 - 2 \cdot 3 \cdot 5 \cdot 0.6 \\ &= 34 - 18 \\ &= 16 \\ \therefore x &= 4 \end{aligned}$$

$$\frac{30}{0.5} = 60$$

Figure 8

Calculate the length of RT.

(5 marks)

Continued →

12. a. Solve the equation $x^2 + 5x - 1 = 0$. Correct your answer to 2 decimal places.

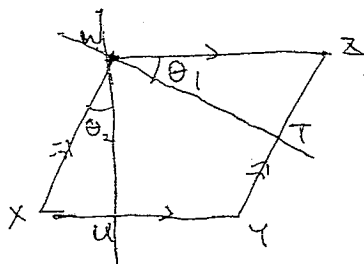
$$x = \frac{-5 \pm \sqrt{25+4}}{2} = \frac{-5 \pm \sqrt{29}}{2}$$

(7 marks)

- b. WXYZ is a parallelogram. A line through W meets ZY at T and XY produced at U.

- (i) Prove that the triangles WZT and UWX are similar.
 (ii) Given that $ZT : TY = 3 : 2$ and the area of triangle WZT = 9 cm^2 , calculate the area of triangle UWX.

(8 marks)



We can't solve, because $\theta_1 \neq \theta_2$. ~~the points T, U aren't located exactly.~~

END OF QUESTION PAPER

NB: This paper contains 6 pages.